Lower degree and fewer witnesses for Diophantine consistency

An explicit strong-trace reduction, integral radical projections, and the diagonal power t^t

AI-assisted mathematical research
Prepared for Vladimir Reshetnikov and the ProveIt project

25 October 2026

Abstract

We give two complementary improvements to fixed Diophantine equations whose unsolvability is provably equivalent in ZFC to Con(ZFC). The compact construction has ten strictly positive unknowns, exact degree 232, and an 87-operation arithmetic circuit; its matched predecessor has degree 304 in 87 operations. Four further compact tradeoffs are supplied. The key observation is that the doubled strong Pell trace is already a polynomial congruent to one modulo the main ordinate. An auxiliary expression can therefore use this trace once instead of twice. The proof treats negative intermediate ranks before invoking their intended interpretations.

For unrestricted integer unknowns, we give completely expanded quartics in 234, 235, and 236 variables. The sparse 236-variable endpoint has 172 quadratic residuals, 4,992 collected monomials, and coefficient height 10,722. A common radical of the quadratic parts exposes integral directions that can be eliminated without increasing degree; six exact projections are certified in each direct endpoint. A further semantic simplification proves that $F = t^t$ suffices for proof-label recovery in the factorial-free binomial compiler.

All resulting equations are ordinary polynomials with fixed integer coefficients. Complete literal data, exact transformations, executable checks, retained arithmetic proofs, and source calibrations accompany the paper. These are ordinary mathematical proofs composed with identified repository results; no new end-to-end Lean/Rocq or encoded object-ZFC proof is claimed. The comparisons concern an immutable inspected snapshot, and do not establish a globally simplest equation or literature-wide priority.

Source snapshot: `b7e35224c04fe1031c0a53f80a395e9b32205022`.
Positive and unrestricted integer domains are kept distinct.

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1 The result and the comparison standard

A small number of unknowns can make a formula short while requiring enormous fixed coefficients and high degree. A direct encoding of finite proof certificates can keep all coefficients small and the degree four, at the cost of more unknowns. We improve both directions without collapsing their different measures into a single unsupported minimum claim.

Our two main displayed equations are

$$P_{\mathrm{tr}}(J, F, \alpha, i, T, w, s, n, \eta, \zeta) = 0, \qquad \boldsymbol{u} \in \mathbb{Z}_{>0}^{10}, \tag{1}$$

$$P_{236}(x_0, \ldots, x_{235}) = \sum_{j=0}^{171} R_j(\boldsymbol{x})^2 = 0, \qquad \boldsymbol{x} \in \mathbb{Z}^{236}. \tag{2}$$

Section 2 defines every polynomial abbreviation in the first equation. Appendix A prints every quadratic R_j in the second. The archive also contains complete expansions of all three direct quartics, with ordered coordinates and integer coefficients. The four fixed compact numerals are specified by the retained finite recipe and exact binary payloads; they are not adjustable parameters.

Theorem 1.1 (Consistency equivalences)**.** *For the literal equations supplied in this release, the new proofs below, composed with the identified inherited arithmetic and syntactic calibration, establish*

$$\mathrm{ZFC} \vdash [\neg\exists \boldsymbol{u} \in \mathbb{Z}_{>0}^{10}\ P_{\mathrm{tr}}(\boldsymbol{u}) = 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}),$$
$$\mathrm{ZFC} \vdash [\neg\exists \boldsymbol{x} \in \mathbb{Z}^{d}\ P_{d}(\boldsymbol{x}) = 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}), \quad d \in \{234, 235, 236\}.$$

The same positive-domain statement holds for the other four supplied compact circuits. No consistency assumption is used in deriving these equivalences.

This is an equivalence provable in ZFC, not a proof in ZFC of either side. Each equation has a solution precisely when a finite refutation exists, under its stated encoding. Finite experiments with candidate solutions would not establish consistency.

1.1 Measured changes

Unknowns/domain	Operations	Parent degree	New degree
10 positive	75	626	402
10 positive	76	464	352
10 positive	79	382	290
10 positive	80	338	258
10 positive	87	304	232

Table 1: The five compact improvements. Each retains its parent's operation count and complete fixed coefficient tuple.

Integer quartic	Variables	Rows	Terms	Height
Sparse diagonal power	236	172	4,992	10,722
Dense diagonal power	235	171	10,536	10,730
Shared factorial error	234	171	21,385	44,800

Table 2: Three new direct equations over unrestricted integers. All have exact degree four.

The matched direct parents have respectively 242, 241, and 240 variables. The first two new branches also replace the label base by t^t. Each final radical projection phase removes exactly six variables and six rows. The 240-variable shared-error parent has 21,482 terms and height 6,144; its 234-variable descendant reduces dimension and support but increases height.

Every operation count charges a live binary addition, subtraction, or multiplication, including a square and multiplication by a fixed scalar. Repeatedly used named values are shared. Such names are not new existential unknowns. Degree means total degree after all abbreviations are substituted, and the finite verifier certifies an attained highest homogeneous form, not merely a syntactic upper bound. A direct polynomial's term count refers to its fully collected sum of squares; its coefficient height is the largest absolute coefficient in that expansion.

The compact results keep all four old numerical ports intact. Their degree and evaluation improvements therefore do not hide a coefficient change. The direct results expose their support/height tradeoffs. The 234-variable endpoint saves the most coordinates among the three direct forms, while 236 is much sparser. Neither comparison changes the domain.

2 The compact equation

All ten independent variables in (1) are strictly positive. Let β, K_0, M_C, M_F be the complete inherited BFS943 compact44 tuple, with its corrected initializer and source convention retained. Its fixed numerical bounds include

$$\beta \geq 31 \text{ odd}, \qquad K_0 > 0, \qquad 0 < M_C < \beta < M_F < 2\beta.$$

These bounds help prove the arithmetic reduction, but do not alone select the ZFC instance. The full fixed tuple in the archive selects that instance. Define

$$\begin{gathered}
h = \beta J, \qquad q = h + 1, \qquad C = q - F - \alpha, \\
X = wq, \qquad Y = qs, \qquad k = \eta + \zeta, \qquad c = kY + \eta, \\
A = Y(X + 1) + 1, \qquad \Delta = A^2 - 1, \qquad H = 4A - 5, \\
v = XY^2, \qquad E = 1 + v(v + 1)k^2, \\
R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
B = (K_0 + w)C - F, \qquad U = ic^2, \\
a = \Delta U^2 E, \qquad G = E(1 + a), \qquad f = 1 + 2a, \\
Q = 4\Delta^2 U^2 G, \qquad b = cT - R, \qquad V = fb - c, \\
L = bV, \qquad K = X(3c - 2X) + 2, \\
e_6 = 6(k - R - 1) - v(R + 1)(R - 1)(R + 3), \\
D_6 = 6q^6, \qquad M = hD_6 H, \\
Z = hD_6[H(n - LB) + K] - H[D_6 B + h(e_6 - 6q^4 s)].
\end{gathered}$$

The promised polynomial is

$$\boxed{P_{\mathrm{tr}} = Q(Z^2 - M^2 G V^2) + M^2 G - Z^2.} \tag{3}$$

All these expressions are ordinary polynomial abbreviations. The verified circuit evaluates exactly (3), exploiting $Q = 4\Delta a(1 + a)$ and the shared products listed in its JSON definition. Its exact count is 87, and its exact total degree is 232.

The lower-case f is the explicit strong Pell trace. It differs from the independent uppercase variable F in the coefficient mask. Neither should be identified with the $F = t^t$ variable in the separate direct construction.

2.1 The algebraic opportunity

The matched predecessor sets $V = f^2b - c$, evaluating f^2 through the identity $1 + 4a(1 + a)$. We instead set

$$V = f(cT - R) - c. \tag{4}$$

Since $U = ic^2$, we have $c \mid a$ and therefore

$$f \equiv 1 \pmod{c}, \qquad V \equiv -c \pmod{f}, \qquad V \equiv -R \pmod{c}. \tag{5}$$

The old proof needs precisely the latter two residues at its index-recovery stage. The special doubled-trace formula supplies $f \equiv 1 \pmod{c}$, whereas the previous expression only exploited $f^2 \equiv 1 \pmod{c}$.

A congruence observation alone would be insufficient. The proof also needs absolute size estimates and positive witness construction. Some intermediates can be negative at arbitrary positive inputs. The following sections verify these obligations for the changed expression.

3 Soundness of the trace reduction

We retain the proved Pell folding and native decoding interfaces from the identified source packets, and state where they enter. Their supporting proofs and calibrations are included in the release. We use

$$\chi_A(m) + \psi_A(m)\sqrt{A^2 - 1} = (A + \sqrt{A^2 - 1})^m \qquad (m \geq 0).$$

3.1 Recover roots before interpreting the mask

Positive inputs imply $q \geq 32$, $X, Y \geq q$, $k \geq 2$, $c > kY$, $A \geq 3$, $U \geq c^2$, $a > 0$, $f > 2c$, $Q > 1$, and $M > 0$. For integral b, the value $V = fb - c$ is nonzero: if $b = 0$ then $V = -c$; otherwise $|V| \geq f - c > 1$. At a zero of (3), write $z = Z/M$ to obtain

$$(Q - 1)z^2 = G(QV^2 - 1). \tag{6}$$

Since $Q = 4\Delta^2U^2G$, both $Q - 1$ and $QV^2 - 1$ are coprime to G. Prime valuations force each exponent in G to be even. Thus $G = g^2$ with $g > 0$. Moreover $\gcd(E, 1 + a) = 1$, because $E \mid a$, so $E = \tau^2$, $1 + a = f_0^2$, and $g = \tau f_0$ for positive integers.

The inherited rational adjacent-Pell descent, applied to $QV^2 - (Q - 1)y^2 = 1$ with $y = |z|/g$, proves $y \in \mathbb{Z}_{>0}$. This is an integrality theorem, not an assumption that a rational square root is integral. Thus $z = \varepsilon gy \in \mathbb{Z}$ for $\varepsilon \in \{-1, 1\}$. Put $I = 2Ug$. The exact identities

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad c^2 \mid I \tag{7}$$

are unchanged. The signed folding interface uses (7), the adjacent norm, $c \leq I$, and $V \equiv -c \pmod{f}$; it does not use the extra trace factor in the old V. It yields $c = \psi_A(p)$ for a positive odd p. The first norm gives $k = 2\psi_{2v+1}(m)$ for $m \geq 1$. Index $m = 1$ would force $k = 2$, $\eta = \zeta = 1$, and $c = 2Y + 1$, strictly between the first two positive main ordinates. Hence $m \geq 2$ and

$$c > kY \geq (8v + 4)Y > 8XY^3 \geq 8q^4. \tag{8}$$

In particular $c > X$ and $K > 0$. No positivity of R, C, B, b, or V has been assumed.

3.2 The absolute estimate needed for transport

If $b = 0$, the adjacent norm would give $y^2 - c^2 = (c^2 - 1)/(Q - 1)$ strictly between zero and one. Here $Q = (\Delta I)^2 \geq 9c^4 > c^2$ by (7). Thus $b \neq 0$, and b, V have the same sign. With $D_0 = \Delta U^2$,

$$\frac{a^2}{g^2} = \frac{D_0 a}{1 + a} > \frac{D_0}{2} > 9.$$

Consequently $a > 3g$, $|V| \geq f - c > a > 3g$, $|V| > |b|$, and $y < 3|V|/2$.

The exclusion of main index $p = 3$ is independent of the formula for V. If $c = 4A^2 - 1$ and $m \geq 3$, Pell growth gives $k > 32v^2$, incompatible with $c = kY + \eta$. Thus $m = 2$ and $k = 8XY^2 + 4$. For $u = X + 2 - 2Y$, expansion gives

$$\eta = 4Y^2(Xu + 1) + 8Y(X + 1) + 3 - 4Y.$$

If $u \leq -1$ this is negative; if $u \geq 2$ it exceeds k. Both contradict $0 < \eta < k$. The remaining cases force $2Y - X \in \{1, 2\}$, impossible since $q \mid X, Y$ and $q \geq 32$. Hence $c > c_3 := 4\Delta + 3$.

Classification of the adjacent norm, with $S = \Delta I$, gives $|V| = \chi_S(j)/S$ for odd $j \geq 1$. Index one is excluded. Index three would give $|V| = 4Q - 3$. As $Q = \Delta(f^2 - 1)$, either sign would imply one of

$$f(b - 4\Delta f) = c - c_3, \qquad f(b + 4\Delta f) = c + c_3.$$

The right sides are strictly between zero and $2c < f$. Thus $j \geq 5$ and

$$|V| \geq 16Q^2 - 20Q + 5 > 8Q^2, \qquad |b| > \frac{8Q^2 - c}{f} > 8f^3 - \frac{1}{2} > 4g.$$

Here $Q > f^2$ follows from $\Delta \geq 3$ and $f \geq 2$. Therefore $L = bV = |b||V|$ satisfies

$$L > 0, \qquad gy < \frac{3}{8}L < \frac{1}{2}L, \qquad \frac{|b|}{q^6} < \frac{1}{2}L. \tag{9}$$

3.3 Negative ranks in the Lucas-corrected equation

For proof only, put

$$t_0 = B/h, \quad W = 1 + hL, \quad \kappa = K/H, \quad \sigma = s/q^2, \quad \bar{e} = e_6/6, \quad \rho = \bar{e}/q^6.$$

The carrier identity gives

$$n = \varepsilon g y - \kappa - \sigma + W t_0 + \rho. \tag{10}$$

Initially $\bar{e}$ may be rational. Write $B_R = (R + 1)(R - 1)(R + 3)$. Since $cT > k - 1$,

$$\rho < \frac{b}{q^6} - \frac{v B_R}{6q^6}.$$

If $B_R \geq 0$ and $B \leq -1$, then $Wt_0 < -L$; positivity of κ, σ and (9) force $n < 0$.

The adverse sign $B_R < 0$ occurs precisely at integer ranks $R \leq -4$ and $R = 0$. For $R = -r \leq -4$, the exact mask identity

$$R = (hC + q\alpha)(q^2 - 1) + (M_C + qM_F)J$$

gives $C < 0$, $|C| > r/q^3$, and $B \leq -1 - w|C|$. The adverse correction is less than $ws^2r^3/(6q^3)$. But $b = cT + r > r, c$, so $V > (f - 1)b$ and $L > (f - 1)r^2$, while $f - 1 = 2a > s^2$. The additional term

$-w|C|L$ therefore dominates this correction. The remaining terms again yield $n < 0$ by (9). For $R = 0$, we have $|C| \geq 1$ and $L > f - 1 > s^2$; the adverse term $ws^2/(2q^3)$ is likewise dominated. This covers every integer rank.

It follows that $B \geq 0$ and $C > 0$, because the independent F is positive. The inherited mask estimate is now licensed: $3q < R < q^4 - 2$. Equation (8) gives $c > 2R$, so $b, V > 0$. The residues (5) and the strong pair (7) now meet all hypotheses of the normalized rank and signed auxiliary-index lemmas. They yield $p = R$ and the old parity conclusions. The other size premise $c > 2p$ follows from $c = \psi_A(p)$, $A \geq 3$, and the already established $p \geq 5$.

3.4 The unchanged decoding stages

The identity

$$2\psi_{1+2v}(m) = 2m + 8v\binom{m+1}{3} + 32v^2 Z_m, \qquad Z_m \in \mathbb{N},$$

and the carrier recover the first index and remaining denominators. Their data q, X, Y, k, c, R and integral carrier z have not changed. The inherited proof first obtains divisibility by q^4, bounds the index difference by $R < q^4$, and only then obtains divisibility by q^6. Its valuation calculation covers the primes two and three dividing the factor six. The whole-product projection, native witnesses, and fixed-source refutation calibration then apply. The retained source proof supplies these unchanged stages; no new cancellation modulo a possibly nonunit factor is introduced.

4 A polynomial completeness map

There is a concrete map from every old positive zero, stronger than rebuilding canonical witnesses alone. Define the polynomial

$$d = \frac{f-1}{c} = 2\Delta i^2 c^3 E.$$

Given an old positive zero with $V_{\text{old}} = f^2 b_{\text{old}} - c$, set

$$T_{\text{new}} = fT_{\text{old}} - dR. \tag{11}$$

All other independent inputs except n are fixed. Then identically

$$b_{\text{new}} = fb_{\text{old}}, \qquad V_{\text{new}} = V_{\text{old}}.$$

For the $L = bV$ charts, set

$$n_{\text{new}} = n_{\text{old}} + (f-1)L_{\text{old}}B.$$

For the $L = V^2$ chart, leave n unchanged. In either case $n_{\text{new}} - L_{\text{new}}B = n_{\text{old}} - L_{\text{old}}B$, so the entire carrier and final polynomial are preserved under this substitution.

At an old positive zero, the old proof gives $0 < R < c/2$ and $B \geq 0$. Since $T_{\text{old}} \geq 1$, $T_{\text{new}} > f-(f-1)/2 > 0$; it is integral because d is a polynomial. The old b, V, L are positive, so $n_{\text{new}} > 0$ as well. This proves completeness on the required domain. Soundness for arbitrary new positive zeros was proved separately above. We do not claim an integral inverse of this map on every new zero.

The same trace replacement works in four simpler supplied charts. Their complete formulas and inherited source proofs are retained in the compact component; the altered transport estimates are no stronger than those just established. Table 1 records all five degree/cost tradeoffs.

5 A smaller arithmetic certificate base: $F = t^t$

This section belongs to the direct proof-calculus construction and uses a different F from the compact mask. The factorial-free predecessor uses $F = [2(t+1)]^t$ to separate proof-label multiplicities and make its binomial CRT modulus cover every row. The diagonal power $F = t^t$ suffices for both jobs. Its exact power relation is compiled into polynomial Pell residuals; it is not left in the final Diophantine equation.

Lemma 5.1 (Label balance at the diagonal power)**.** *Let $t \geq 1$, $F = t^t$, and $W_j = 1 + jF$ for $0 \leq j \leq t$. Suppose $v_i, p_i, q_i \in \{0, \ldots, t\}$ for $0 \leq i < t$, and*

$$\prod_{i<t} W_{v_i} = (1+F) \prod_{i<t} W_{p_i} W_{q_i}. \tag{12}$$

At every positive label j, its conclusion multiplicity minus its premise multiplicity is one at $j = 1$ and zero otherwise.

Proof. Discard the label-zero factors, which are one. If the two factor counts differ, their smaller count r is at most t, because the left side has at most t nonunit factors. The side with more factors is at least $(F+1)^{r+1}$, and the other side is at most $(1+tF)^r \leq t^r(F+1)^r$. But $F + 1 = t^t + 1 > t^r$, a contradiction. Hence their counts agree, with common value $r \leq t$.

Replace F by an indeterminate z. Each side is a product of r linear factors $1 + jz$, where $1 \leq j \leq t$. Its coefficient of z^k, for $0 \leq k \leq r$, is positive and at most

$$\binom{r}{k} t^k \leq t^{r-k} t^k = t^r \leq F.$$

For the inequality, inject the $(r-k)$-element subsets of an r-element set into ordered $(r-k)$-tuples from a t-element set. Thus every coefficient of the difference polynomial has absolute value strictly less than F. If the difference were nonzero, divide its evaluation at F by the power corresponding to its lowest nonzero coefficient and reduce modulo F. That coefficient would be a nonzero multiple of F of absolute value less than F. This is impossible. The products are therefore equal in $\mathbb{Z}[z]$, and the distinct roots $-1/j$ give the claimed multiplicities. When $F = 1$, the coefficient difference bound itself already implies that the difference is zero. □

The strict bound is on the *difference* coefficients. Bounding each entire product by its coefficient sum misses this improvement and retains the unnecessarily larger base $(t+1)^t$.

Lemma 5.2 (Single-conclusion flow)**.** *Take a finite multiset of valid Horn rules with one conclusion and at most two premises; pad unused premises with freely derivable truth, label zero. If their multiplicities have the balance in Lemma 5.1, then label one has a finite derivation.*

Proof. Form the closure of truth under these finitely many rule instances. If label one were outside, every rule whose conclusion is outside would have at least one premise outside. The sum of conclusions minus premises over outside labels would be at most zero. The prescribed balance instead gives one, a contradiction. □

This inherited flow argument explains why an unordered certificate with cycles does not create a false proof. It needs no ordering of the rows.

5.1 The binomial compiler still covers every row

Let $\Lambda_t = \operatorname{lcm}(1, \dots, t)$ and $\mathcal{R} = F^2$. We need only *numerical domination*, not $\Lambda_t \mid F$:

$$\Lambda_t \le t! \le t^t = F, \qquad \mathcal{R} \ge \Lambda_t^2.$$

Indeed Λ_t need not divide t^t: three divides Λ_{16} but not 16^{16}. Avoiding this divisibility requirement was the purpose of the inherited factorial-free compiler.

For clarity, its two prime-power facts are recalled here. Put $\mathcal{B} = \binom{\mathcal{R}+t}{t}$. Every full prime power $p^e \| \mathcal{B}$ divides some $\mathcal{R} + j$, $1 \le j \le t$: Legendre's valuation formula consists of zero-one summands and its sum is at most $\max_j v_p(\mathcal{R} + j)$. Conversely, each row j has a full prime power of $\mathcal{B}$ exceeding t. Since $\mathcal{R} + j > \Lambda_t^2$, some $v = v_p(\mathcal{R} + j)$ exceeds $2\lfloor \log_p t \rfloor$. For this prime, the valuation in $\mathcal{B}$ is exactly

$$v - v_p\left(t\binom{t-1}{j-1}\right).$$

The subtracted valuation is at most $\lfloor \log_p t \rfloor$, because $t\binom{t-1}{j-1} \mid \Lambda_t$. This classical denominator bound also follows from Farhi's binomial lcm identity [5]. The resulting full prime power therefore exceeds t and identifies the row uniquely. The first fact covers *every prime* in the converse CRT construction, while the second covers *every row* in soundness. Both remain valid with $F = t^t$.

5.2 Parity and cofinal completeness

The retained horizon is $t = 16 + h_0^2$. For completeness choose the cofinal subfamily with even h_0, so t is even. Then

$$v_2(F) = t\, v_2(t) \ge t.$$

Thus F and the beta base $y = F(2u+1)^2$ are even. Let O be the odd part of $t!$. Choose $2u+1$ to be a sufficiently large positive odd multiple of O. Since $v_2(F) \ge t > v_2(t!)$, this makes y an even multiple of $t!$, as required by the inherited pairwise-coprime beta-modulus construction. This choice supplies the missing odd prime factors; it does not incorrectly assume $t! \mid t^t$. The beta moduli $1 + (j+1)y$ are all odd, so their CRT progression admits arbitrarily large even representatives for the beta code. The binomial modulus $\mathcal{B}$ is odd: since $v_2(F^2) > \lfloor \log_2 t \rfloor$, $v_2(F^2 + j) = v_2(j)$ for $1 \le j \le t$, and the numerator and $t!$ have equal two-adic valuations. All inherited CRT parity choices remain available.

Soundness imposes no new evenness condition on an arbitrary satisfying horizon. The independent audit examines each use of the old base $[2(t+1)]^t$, rather than replacing it in the formula and assuming all estimates survive. The needed size bound follows already from $F = t^t \ge 16^{16} = 2^{64}$ at $t \ge 16$, together with $F \ge \Lambda_t$. The source edit also changes the actual power-call base and the shared frame metadata, which are checked separately.

6 Eliminating common quadratic directions

The second direct improvement is algebraic and reusable. An ordinary monic substitution can raise degree: replacing a variable by a quadratic expression where it appears quadratically elsewhere may create quartic residuals and an eighth-degree sum of squares. Instead, we find directions along which *every* residual's quadratic part is constant.

Write

$$R_j(\boldsymbol{x}) = c_j + \ell_j(\boldsymbol{x}) + Q_j(\boldsymbol{x}),$$

where ℓ_j and Q_j are homogeneous of degrees one and two. Let H_j be the integer Hessian matrix of Q_j; a term ax_r^2 contributes $2a$ in its diagonal position.

Definition 6.1. The common quadratic radical is

$$\mathcal{K} = \bigcap_j \ker_{\mathbb{Q}} H_j \subseteq \mathbb{Q}^n.$$

Theorem 6.2 (Integral radical projection)**.** *Suppose $\boldsymbol{d} \in \mathbb{Z}^n \cap \mathcal{K}$ has $d_v = 1$ at a coordinate v, and $\ell_r(\boldsymbol{d}) = \varepsilon \in \{1, -1\}$ for a residual r. The common integer zero set of the system is then in explicit polynomial bijection with the common integer zero set of $m - 1$ quadratic residuals in $n - 1$ coordinates.*

Proof. Make the integral coordinate change

$$x_v = y_v, \qquad x_w = y_w + d_w y_v \quad (w \neq v).$$

Its inverse is $y_v = x_v$, $y_w = x_w - d_w x_v$, so it is unimodular. Since $H_j \boldsymbol{d} = 0$, the quadratic part of every transformed residual is independent of y_v. Each transformed residual is

$$S_j(\widehat{\boldsymbol{y}}) + \ell_j(\boldsymbol{d}) y_v,$$

where the hat omits coordinate v and S_j is quadratic. The selected row has the unique integral solution $y_v = -\varepsilon S_r$. Substituting in all other rows gives

$$S_j - \varepsilon \ell_j(\boldsymbol{d}) S_r,$$

still quadratic. Delete the selected row and coordinate. Restriction maps an old zero to a new zero. Restoring y_v by the displayed expression and undoing the shear gives its unique inverse. All maps have integer coefficients. □

The sum of squares still has degree at most four. Its polynomial need not equal the old polynomial; the zero sets correspond. This distinction matters when comparing collected term counts and coefficient heights.

6.1 A small example

For

$$R_1 = (x + y)^2 + z - 1, \qquad R_2 = (x + y)^2 + x + 2y,$$

the direction $(1, -1, 0)$ belongs to both Hessian kernels. Substituting $x = u$, $y = v - u$ gives $R_1 = v^2 + z - 1$ and $R_2 = v^2 + 2v - u$. The second row restores $u = v^2 + 2v$, leaving only $v^2 + z - 1 = 0$. Both the shear and the restoration are integral. The large certificates use exactly this mechanism.

6.2 The six actual projections

At every stage, the supplied certificate records the ordered variable names, the integer radical direction, the unit coordinate, the selected row, the shears, the quadratic restoration, and the retained-coordinate map. Each branch uses six such projections. Each step deletes exactly its selected coordinate and row; no further unused variable is discarded at this stage.

The independent checker verifies $H_j \boldsymbol{d} = 0$ for every row, the unit slope, both coordinate identities, the restored pivot equation, and every substituted residual as an exact polynomial identity. It expands and

collects the final sum of squares. Earlier preprocessing may remove unused coordinates; that separate existence-of-zeros step is not described as a bijection of all raw assignments.

The search computes rational kernels and clears denominators, but its output certificate is integral. No search heuristic is a mathematical premise. Exact modular rank certificates show that the final stacked Hessian matrix has full column rank modulo 101 in all three branches. The corresponding integer minor is nonzero, so the common radical is zero over $\mathbb{Q}$. There are no further directions of this particular type at these charts. This is not a global minimality statement.

7 From proof calculus to the fixed integer quartics

The direct branch uses the numerical proof calculus C_{222}. The retained source compiles its finite numerical Horn rule families, an unordered flow certificate, bounded range assertions, binomial CRT row selection, and a synchronized Pell power representation. The result is a system of quadratic residuals with fixed integer coefficients. It has no Turing machine coefficient tuple and does not invoke Jones's universal polynomial.

The 235- and 236-variable branches begin with the factorial-free compiler [3], change the power request to t^t by Section 5, and apply the six radical projections. Their support tradeoff reflects an earlier dense affine elimination used only in the 235 branch. The 234-variable branch applies the six projections to the shared-factorial-error quartic [4]; it does not claim to remove that branch's factorial identity.

The relevant chain, formalizable internally in ZFC, is

$$\begin{aligned} &\text{an integer zero of the final quartic} \\ \iff &\text{a valid compiled arithmetic certificate} \\ \iff &\text{a finite } C_{222} \text{ refutation} \\ \iff &\neg\,\mathrm{Con}(\mathrm{ZF}) \iff \neg\,\mathrm{Con}(\mathrm{ZFC}). \end{aligned}$$

The last step is syntactic equiconsistency, using the retained constructible-model calibration. It does not assume a transitive model or an externally standard model. The archived calibration review identifies the source interfaces on both sides.

An external proof-assistant theorem about standard numerical objects and an encoded object-language ZFC derivation are different artifacts. This release supplies ordinary arguments formalizable in ZFC and retains the inherited formal-interface sources. It does not claim that every stage has been newly elaborated as a Lean, Rocq, or object-ZFC proof term. A finite Python replay authenticates coefficients and transformations; it does not replace unbounded Pell, CRT, and syntactic calibration arguments.

8 Verification and reproducibility

The final release checks are organized by mathematical obligation.

- The compact checker authenticates 105 inherited source files and all four fixed numerical payloads. For each of the five circuits it checks topology, liveness, the complete parent/child gate relation, and every polynomial stage through 32–37 exact cuts. Nine auxiliary identities include the strong-trace algebra and polynomial completeness map.
- It computes the complete highest homogeneous forms. Eighteen deliberate operator corruptions per circuit are rejected, for a total of 90. Normal and optimized-Python receipts agree. Exact reconstruction of the fixed tuple and byte-for-byte circuit rebuilding pass.

- The direct checker independently represents monomials as variable/exponent pairs. It checks the actual diagonal-power source edit, 18 Horn identities, 202 predecessor monic projections, the selected affine maps, and all six new radical projections for each endpoint.
- It independently expands every final quartic, verifies every coefficient, checks all reported metrics, and verifies the terminal Hessian ranks modulo 101. Five deliberate corruptions of directions, restorations, coordinate deletion, residuals, and expanded coefficients are rejected.
- The scoped predecessor checks pass both normally and with Python assertions disabled. A disposable reconstruction reproduces 28 selected mathematical JSON files byte for byte. These retained dependency subsets omit historical PDFs and obsolete experiments; their scoped entry points are explicitly documented.
- A root release replay checks the component verifiers and ensures that the report's literal residual appendix, comparison tables, and degree examples match their canonical data. The PDF is rendered and visually inspected before packaging.

The release uses Python 3.10 or newer and its standard library for these exact checks. Run from the extracted package directory:

```
python3 -B verify.py
python3 -B verify.py --rebuild
sh build_report.sh
```

The first command checks the frozen artifacts without changing them. The optional second additionally regenerates the selected artifacts in disposable locations. The last rebuilds the paper from its TeX source. Internal independent mathematical reviews are retained in `review/`; they are AI-assisted reviews, not external human peer review.

The canonical equations use ordered sparse terms: $[a, [i_1, \ldots, i_r]]$ denotes $ax_{i_1} \cdots x_{i_r}$. Repeated indices are repeated factors, and an empty index list is a constant. Every array is finite. The evaluator accepts an ordered tuple of exact integers; it is not a solver or a consistency decision procedure.

Every compact coefficient is included with its inherited exact reconstruction. The actual native tuple, not an arbitrary tuple satisfying a few bounds, is bound to the consistency statement. Checks distinguish coefficient reconstruction, literal circuit identity, degree certification, expansion, transformation correctness, and the unbounded mathematical proof.

The source snapshot and archive hashes are in the ledger. The final manifest authenticates the deliverable bytes. The TeX source builds with a standard TeX Live installation; font versions and PDF metadata can change the PDF hash without altering the canonical equations.

9 What this iteration adds, and further research

The compact advance is the use of the explicit strong-trace congruence $f \equiv 1 \pmod{c}$, together with a complete signed proof, to replace $f^2(cT - R) - c$ by $f(cT - R) - c$. This lowers degree at the same binary operation count: the new trace is evaluated as $f = a + (1 + a)$ using the already shared $1 + a$. The new polynomial parent-to-child map gives a further algebraic check on completeness.

The direct advance is a reusable integral projection mechanism that finds six hidden removable directions. The t^t lemma separately simplifies an actual power request and its justification. The projection applies to more than one inherited source, and the label lemma is proved before any quadratic projection.

The following questions are concrete directions for subsequent work. No unproved answer is used in this release.

1. Can a weaker auxiliary congruence yield a further compact trace deletion? A new index argument would be needed: the present residue modulo f does essential work.
2. Can a changed residual presentation expose more removable directions after the common radical becomes zero? Constant invertible row operations preserve the common radical, so they cannot suffice alone.
3. Can a structured nonlinear coordinate change preserve quadratic residuals and expose more unit coordinates? A valid proposal needs an integral inverse or an explicit existential restoration.
4. Can a simpler proof calculus reduce the direct front end enough to compensate for its arithmetic overhead and calibration? Counting rule schemata alone does not measure the final polynomial.
5. Can the synchronized power compiler use the diagonal nature of $F = t^t$ more deeply than changing its base? Soundness must still cover arbitrary integer assignments.
6. Can the radical-projection and trace-congruence lemmas first be formalized as reusable components, then connected to the internal-ZFC calibration? This would address the actual formal gap rather than recheck only finite numerical cases.

A Complete residuals of the sparse integer equation

Every $x_0, \dots, x_{235}$ below ranges over $\mathbb{Z}$. The equation is exactly $\sum_{j=0}^{171} R_j^2 = 0$. This appendix is generated from the canonical residual JSON and checked against those literal coefficients. The other two residual lists and all three collected quartics are in the archive.

$$
\begin{aligned}
R_{0} &= -x_{3}^{2} + x_{4} \\
R_{1} &= x_{0} - x_{0}\,x_{2} - x_{2} + x_{8} \\
R_{2} &= -x_{2}\,x_{6} - x_{3} + x_{7}\,x_{8} \\
R_{3} &= -1 + x_{1} - x_{7}\,x_{8} \\
R_{4} &= 1 + x_{4} - x_{5}\,x_{19} + x_{17} \\
R_{5} &= -1 - x_{9}^{2} - 2\,x_{9}\,x_{10} - x_{10}^{2} + x_{20} \\
R_{6} &= -10\,x_{9} + x_{9}^{2} - 6\,x_{9}\,x_{10} - 8\,x_{9}\,x_{20} - 20\,x_{10} - 11\,x_{10}^{2} - 16\,x_{10}\,x_{20} - 15\,x_{20} - 4\,x_{20}^{2} \\
&\quad + x_{21} \\
R_{7} &= -2 + 2\,x_{9} - x_{9}^{2} - 4\,x_{9}\,x_{21} - 4\,x_{21} - 4\,x_{21}^{2} + x_{22} \\
R_{8} &= 1 + 6\,x_{9} - 36\,x_{9}^{2} - 24\,x_{9}\,x_{22} + 2\,x_{22} - 4\,x_{22}^{2} + x_{23} \\
R_{9} &= -2 - x_{10} - x_{10}^{2} - 2\,x_{10}\,x_{11} - x_{11} - x_{11}^{2} + x_{24} \\
R_{10} &= -1 + x_{9} - x_{9}^{2} - 2\,x_{9}\,x_{11} + x_{11} - x_{11}^{2} + x_{25} \\
R_{11} &= -16 + 14\,x_{9} - 2\,x_{9}^{2} + 8\,x_{9}\,x_{11} - 8\,x_{9}\,x_{24} - 8\,x_{11} + 8\,x_{24} - 4\,x_{26}^{2} + x_{27} - 2\,x_{44} \\
R_{12} &= -2 - 8\,x_{9} + 8\,x_{9}\,x_{24} + 8\,x_{9}\,x_{25} - 8\,x_{10} + 8\,x_{10}\,x_{24} + 8\,x_{10}\,x_{25} - 4\,x_{20} + 8\,x_{24} - 4\,x_{24}^{2} \\
&\quad - 8\,x_{24}\,x_{25} + 8\,x_{25} - 4\,x_{25}^{2} + x_{28} \\
R_{13} &= 6 + 4\,x_{9} + 32\,x_{9}\,x_{10} + 16\,x_{9}\,x_{20} + 16\,x_{9}\,x_{25} + 8\,x_{9}\,x_{28} + 24\,x_{10} - 32\,x_{10}\,x_{25} - 16\,x_{10}\,x_{28} \\
&\quad + 12\,x_{20} - 16\,x_{20}\,x_{25} - 8\,x_{20}\,x_{28} - x_{21} + 8\,x_{25} - 16\,x_{25}^{2} - 16\,x_{25}\,x_{28} + 4\,x_{28} - 4\,x_{28}^{2} \\
&\quad + x_{29} \\
R_{14} &= 7 - 28\,x_{9} - 8\,x_{9}\,x_{11} + 16\,x_{9}\,x_{20} + 8\,x_{9}\,x_{25} - 27\,x_{10} - 9\,x_{10}^{2} + 20\,x_{10}\,x_{11} - 16\,x_{10}\,x_{24} \\
&\quad + 12\,x_{10}\,x_{25} - 32\,x_{11} - 28\,x_{11}^{2} + 24\,x_{11}\,x_{20} - 24\,x_{11}\,x_{24} + 8\,x_{20}\,x_{25} + x_{21} - 32\,x_{24} \\
&\quad - 4\,x_{24}^{2} - x_{26} + x_{30} + x_{44} \\
R_{15} &= 34 - 2\,x_{9} - 8\,x_{9}^{2} - 16\,x_{9}\,x_{10} - 34\,x_{9}\,x_{11} - 16\,x_{9}\,x_{20} - 4\,x_{9}\,x_{24} - 24\,x_{9}\,x_{25} + 22\,x_{10} \\
&\quad + 14\,x_{10}^{2} - 36\,x_{10}\,x_{11} - 16\,x_{10}\,x_{25} - 15\,x_{11} - 13\,x_{11}^{2} - 24\,x_{11}\,x_{20} - 24\,x_{11}\,x_{25} - 8\,x_{20}\,x_{25} \\
&\quad - x_{21} - 2\,x_{24} - 17\,x_{25} - 4\,x_{25}^{2} + x_{26} - x_{44}
\end{aligned}
$$

$$R_{16} = 32 + 46\,x_{9} - 2\,x_{9}^{2} - 16\,x_{9}\,x_{10} - 8\,x_{9}\,x_{11} - 32\,x_{9}\,x_{20} - 8\,x_{9}\,x_{24} - 16\,x_{9}\,x_{25} + 38\,x_{10} + 18\,x_{10}^{2} - 72\,x_{10}\,x_{11} - 24\,x_{10}\,x_{25} + 24\,x_{11} + 16\,x_{11}^{2} - 48\,x_{11}\,x_{20} - 16\,x_{20}\,x_{25} - 2\,x_{21} + 8\,x_{24} + 2\,x_{26} - 2\,x_{30} - 4\,x_{30}^{2} + x_{31} - 2\,x_{44}$$

$$R_{17} = -19\,x_{9} + 8\,x_{9}^{2} + 8\,x_{9}\,x_{20} + 8\,x_{9}\,x_{24} - 46\,x_{10} - 24\,x_{10}^{2} + 24\,x_{10}\,x_{20} + 24\,x_{10}\,x_{24} - 2\,x_{11} + 4\,x_{20} - 4\,x_{20}^{2} - 8\,x_{20}\,x_{24} + 14\,x_{24} - 4\,x_{24}^{2} - x_{25} + x_{32}$$

$$R_{18} = -3 - 6\,x_{9} - 4\,x_{11} - 2\,x_{25} - 4\,x_{32} - 4\,x_{32}^{2} + x_{33}$$

$$R_{19} = 8 + 42\,x_{9} + 7\,x_{9}^{2} + 28\,x_{9}\,x_{21} + 48\,x_{10} - 32\,x_{10}^{2} - 32\,x_{10}\,x_{20} + 64\,x_{10}\,x_{21} - 40\,x_{20}^{2} + 32\,x_{20}\,x_{21} + 14\,x_{21} - 4\,x_{21}^{2} + x_{34} + 2\,x_{37}$$

$$R_{20} = 1 + 6\,x_{9} - 36\,x_{9}^{2} - 24\,x_{9}\,x_{34} + 2\,x_{34} - 4\,x_{34}^{2} + x_{35}$$

$$R_{21} = -x_{10}^{2} - 4\,x_{10}\,x_{35} - 4\,x_{35}^{2} + x_{36}$$

$$R_{22} = -6 - 2\,x_{9} - 4\,x_{10} - 14\,x_{20} - 16\,x_{20}^{2} + x_{37}$$

$$R_{23} = 24 + 21\,x_{10} - 21\,x_{10}^{2} - 18\,x_{10}\,x_{11} - 24\,x_{10}\,x_{24} - 24\,x_{10}\,x_{37} + 3\,x_{11} - 9\,x_{11}^{2} - 16\,x_{11}\,x_{37} + 3\,x_{24} - 12\,x_{24}^{2} - 8\,x_{24}\,x_{37} + 4\,x_{37} - 4\,x_{37}^{2} + x_{38} + 2\,x_{44}$$

$$R_{24} = 16 - 10\,x_{9} - 20\,x_{9}^{2} - 16\,x_{9}\,x_{11} - 24\,x_{9}\,x_{25} - 24\,x_{9}\,x_{38} - 12\,x_{11} - 16\,x_{11}\,x_{25} - 16\,x_{11}\,x_{38} - 14\,x_{25} - 4\,x_{25}^{2} - 8\,x_{25}\,x_{38} + 4\,x_{38} - 4\,x_{38}^{2} + x_{39}$$

$$R_{25} = -x_{11}^{2} - 2\,x_{11}\,x_{12} - x_{12}^{2} + x_{40}$$

$$R_{26} = -29 + x_{9}^{2} - 4\,x_{9}\,x_{40} - 14\,x_{11} - 2\,x_{11}^{2} - 16\,x_{11}\,x_{24} + 16\,x_{11}\,x_{25} - 8\,x_{11}\,x_{40} - 8\,x_{24} + 12\,x_{24}^{2} - 8\,x_{24}\,x_{25} + 22\,x_{25} - 4\,x_{25}^{2} + 4\,x_{40} - 4\,x_{40}^{2} + x_{41} - 4\,x_{44}$$

$$R_{27} = 4 + 2\,x_{9}^{2} - 8\,x_{9}\,x_{40} - 4\,x_{11}^{2} - 16\,x_{11}\,x_{40} - 4\,x_{25} - 16\,x_{25}^{2} - 32\,x_{25}\,x_{40} - 16\,x_{25}\,x_{41} + 4\,x_{40} - 24\,x_{40}^{2} - 16\,x_{40}\,x_{41} - 4\,x_{41}^{2} + x_{42}$$

$$R_{28} = -16\,x_{9}^{2} + 8\,x_{9}\,x_{11} + 32\,x_{9}\,x_{25} + 16\,x_{9}\,x_{28} + x_{11} - x_{11}^{2} - 8\,x_{11}\,x_{25} - 4\,x_{11}\,x_{28} - 16\,x_{25}^{2} - 16\,x_{25}\,x_{28} - 4\,x_{28}^{2} + x_{43}$$

$$R_{29} = -1 - 8\,x_{11} - 4\,x_{11}^{2} + 8\,x_{11}\,x_{24} + 8\,x_{24} - 4\,x_{24}^{2} + x_{44}$$

$$R_{30} = -1 - 9\,x_{11}^{2} + 24\,x_{11}\,x_{24} + 12\,x_{11}\,x_{44} - 16\,x_{24}^{2} - 16\,x_{24}\,x_{44} - 4\,x_{44}^{2} + x_{45}$$

$$R_{31} = 1 + x_{10} - x_{10}^{2} - 4\,x_{10}\,x_{11} - 4\,x_{10}\,x_{45} - 4\,x_{11}^{2} - 8\,x_{11}\,x_{45} - 4\,x_{45}^{2} + x_{46}$$

$$R_{32} = -1 - 4\,x_{46} - 4\,x_{46}^{2} + x_{47}$$

$$R_{33} = -1 - x_{11}^{2} - 8\,x_{11}\,x_{46} - 4\,x_{11}\,x_{47} - 16\,x_{46}^{2} - 16\,x_{46}\,x_{47} - 4\,x_{47}^{2} + x_{48}$$

$$R_{34} = 2 + x_{10} - x_{10}^{2} - 4\,x_{10}\,x_{11} - 4\,x_{10}\,x_{48} - 4\,x_{11}^{2} - 8\,x_{11}\,x_{48} - 4\,x_{48}^{2} + x_{49}$$

$$R_{35} = -4\,x_{43}^{2} - 8\,x_{43}\,x_{49} - 4\,x_{49}^{2} + x_{50}$$

$$R_{36} = -9 + 7\,x_{9} - x_{9}^{2} - 8\,x_{9}\,x_{43} - 4\,x_{9}\,x_{50} + 24\,x_{43} - 16\,x_{43}^{2} - 16\,x_{43}\,x_{50} + 12\,x_{50} - 4\,x_{50}^{2} + x_{51}$$

$$R_{37} = -1 - 4\,x_{51} - 4\,x_{51}^{2} + x_{52}$$

$$R_{38} = -x_{11}^{2} - 8\,x_{11}\,x_{51} - 4\,x_{11}\,x_{52} - 16\,x_{51}^{2} - 16\,x_{51}\,x_{52} - 4\,x_{52}^{2} + x_{53}$$

$$R_{39} = -x_{9} - x_{9}^{2} - 4\,x_{9}\,x_{11} - 4\,x_{9}\,x_{53} - 4\,x_{11} - 4\,x_{11}^{2} - 8\,x_{11}\,x_{53} - 4\,x_{53} - 4\,x_{53}^{2} + x_{54}$$

$$R_{40} = -10 - 12\,x_{54} - 4\,x_{54}^{2} + x_{55}$$

$$R_{41} = 2 + 4\,x_{11} + 16\,x_{11}\,x_{54} + 8\,x_{11}\,x_{55} - 4\,x_{24} - 16\,x_{24}\,x_{54} - 8\,x_{24}\,x_{55} - x_{44} + 8\,x_{54} - 16\,x_{54}^{2} - 16\,x_{54}\,x_{55} + 4\,x_{55} - 4\,x_{55}^{2} + x_{56}$$

$$R_{42} = -10 + x_{10} - x_{10}^{2} + 8\,x_{10}\,x_{11} - 8\,x_{10}\,x_{24} - 4\,x_{10}\,x_{56} - 16\,x_{11}^{2} + 32\,x_{11}\,x_{24} + 16\,x_{11}\,x_{56} - 16\,x_{24}^{2} - 16\,x_{24}\,x_{56} - 12\,x_{49} - 4\,x_{49}^{2} - 4\,x_{56}^{2} + x_{57}$$

$$R_{43} = 1 + 2\,x_{10} - 2\,x_{10}^{2} + 16\,x_{10}\,x_{11} - 16\,x_{10}\,x_{24} - 8\,x_{10}\,x_{56} - 32\,x_{11}^{2} + 64\,x_{11}\,x_{24} + 32\,x_{11}\,x_{56} - 32\,x_{24}^{2} - 32\,x_{24}\,x_{56} - 16\,x_{49}^{2} - 16\,x_{49}\,x_{57} - 8\,x_{56}^{2} - 4\,x_{57}^{2} + x_{58}$$

$$
\begin{aligned}
R_{44} &= -6\,x_{9} - 3\,x_{9}^{2} - 18\,x_{9}\,x_{10} - 16\,x_{10} - 27\,x_{10}^{2} + 4\,x_{10}\,x_{11} + 3\,x_{11} + 3\,x_{11}^{2} - 4\,x_{11}\,x_{20} \\
&\quad - 19\,x_{20} - 8\,x_{20}^{2} + x_{21} + x_{59} - 4\,x_{72} \\
R_{45} &= 1 - 8\,x_{9} - 4\,x_{9}\,x_{11} + 8\,x_{9}\,x_{59} - 16\,x_{10} - 8\,x_{10}\,x_{11} + 16\,x_{10}\,x_{59} + 2\,x_{11} + 4\,x_{11}\,x_{20} - 4\,x_{11}\,x_{58} \\
&\quad + 8\,x_{20} - 8\,x_{20}\,x_{59} - 4\,x_{58}^{2} + 3\,x_{59} - 4\,x_{59}^{2} + x_{60} \\
R_{46} &= -2 - 12\,x_{9} - 16\,x_{9}\,x_{11} + 16\,x_{9}\,x_{60} - 24\,x_{10} - 32\,x_{10}\,x_{11} + 32\,x_{10}\,x_{60} + 2\,x_{11} + 6\,x_{11}^{2} \\
&\quad + 16\,x_{11}\,x_{20} + 8\,x_{11}\,x_{58} + 12\,x_{20} - 16\,x_{20}\,x_{60} + 8\,x_{58}^{2} - 4\,x_{59} - 2\,x_{60} - 4\,x_{60}^{2} + x_{61} \\
R_{47} &= -2 - 4\,x_{61} - 4\,x_{61}^{2} + x_{62} \\
R_{48} &= -x_{3}\,x_{14} + x_{63} \\
R_{49} &= x_{13} - x_{14} - 2\,x_{23}\,x_{63} - x_{64} \\
R_{50} &= x_{13} - x_{14} - 2\,x_{27}\,x_{63} - x_{65} \\
R_{51} &= -2\,x_{29}\,x_{63} + 2\,x_{42}\,x_{63} - 2\,x_{63} - x_{66} + x_{71} \\
R_{52} &= x_{13} - x_{14} - 2\,x_{31}\,x_{63} - x_{67} \\
R_{53} &= x_{13} - x_{14} - 2\,x_{33}\,x_{63} + 6\,x_{63} - x_{68} \\
R_{54} &= x_{13} - x_{14} - 4\,x_{35}\,x_{63} - 2\,x_{36}\,x_{63} - x_{69} \\
R_{55} &= x_{13} - x_{14} - 2\,x_{39}\,x_{63} - x_{70} \\
R_{56} &= x_{13} - x_{14} - 2\,x_{42}\,x_{63} - 4\,x_{63} - x_{71} \\
R_{57} &= -x_{9}\,x_{10} + x_{9}\,x_{11} + x_{10} + x_{10}\,x_{11} - x_{11} - x_{11}^{2} + x_{72} \\
R_{58} &= 1 + 2\,x_{12} + x_{12}^{2} - x_{72}^{2} + x_{73} \\
R_{59} &= -4\,x_{61}\,x_{63} - 2\,x_{62}\,x_{63} + 2\,x_{63} + x_{74} \\
R_{60} &= -x_{13}^{2} + 2\,x_{13}\,x_{14} + 2\,x_{13}\,x_{74} - x_{14}^{2} - 2\,x_{14}\,x_{74} - x_{73}^{2} - x_{74}^{2} + x_{75} \\
R_{61} &= -1 - x_{3}\,x_{10} + x_{76} \\
R_{62} &= -x_{14}\,x_{76} + x_{78}\,x_{79} + x_{80} \\
R_{63} &= -1 + x_{3} - x_{3}\,x_{9} + x_{77} \\
R_{64} &= -1 + 2\,x_{3} + 2\,x_{3}\,x_{9} + 4\,x_{3}\,x_{10} - 2\,x_{3}\,x_{20} + x_{78} \\
R_{65} &= -x_{13}\,x_{77} + x_{79} \\
R_{66} &= -6\,x_{3}\,x_{14} + x_{13}\,x_{76} + 12\,x_{14} - 8\,x_{14}\,x_{76} - 4\,x_{14}\,x_{77} - x_{14}\,x_{78} + x_{81} \\
R_{67} &= -x_{13}\,x_{64} + x_{14}\,x_{64} + x_{82} \\
R_{68} &= -x_{65}\,x_{82} + x_{83} \\
R_{69} &= -x_{66}\,x_{83} + x_{84} \\
R_{70} &= -x_{67}\,x_{84} + x_{85} \\
R_{71} &= -x_{68}\,x_{85} + x_{86} \\
R_{72} &= -x_{69}\,x_{86} + x_{87} \\
R_{73} &= -x_{70}\,x_{87} + x_{88} \\
R_{74} &= -x_{71}\,x_{88} + x_{89} \\
R_{75} &= -x_{80}\,x_{89} + x_{90} \\
R_{76} &= -x_{81}\,x_{90} + x_{91} \\
R_{77} &= -x_{91}^{2} + x_{92} \\
R_{78} &= -x_{2} - x_{2}\,x_{17} + x_{17} + x_{93} \\
R_{79} &= -x_{15}\,x_{93} + x_{94} \\
R_{80} &= -2\,x_{1}^{2} + 2\,x_{1}\,x_{13} + 2\,x_{1}\,x_{14} + 2\,x_{1}\,x_{94} + 2\,x_{1}\,x_{95} - x_{13}^{2} - 2\,x_{13}\,x_{94} - x_{14}^{2} \\
&\quad - 2\,x_{14}\,x_{95} + x_{18}\,x_{97} - x_{75}\,x_{92} - x_{94}^{2} - x_{95}^{2} \\
R_{81} &= -x_{2}\,x_{16} + x_{16} - x_{16}\,x_{93} + x_{95}
\end{aligned}
$$

$$
\begin{aligned}
R_{82} &= -x_{9} - x_{10} - 4\,x_{20} - 2\,x_{23} - 2\,x_{27} - 2\,x_{29} - 2\,x_{31} - 2\,x_{33} - 4\,x_{35} - 2\,x_{36} - 2\,x_{39} - 2\,x_{42} - 4\,x_{61} \\
&\quad - 2\,x_{62} + x_{96} \\
R_{83} &= -x_{98}\,x_{101} + x_{101} + x_{103} \\
R_{84} &= -x_{5}\,x_{125} + x_{99}\,x_{224} - x_{102} - x_{103}\,x_{125} \\
R_{85} &= x_{104}\,x_{225} - x_{106}\,x_{128} - x_{107} \\
R_{86} &= x_{108}\,x_{225} - x_{109}\,x_{128} - x_{110} \\
R_{87} &= x_{111}\,x_{225} - x_{112}\,x_{128} - x_{113} \\
R_{88} &= x_{114}\,x_{225} - x_{115}\,x_{128} - x_{116} \\
R_{89} &= x_{117}\,x_{226} - x_{119}\,x_{128} - x_{120} \\
R_{90} &= x_{121}\,x_{226} - x_{122}\,x_{128} - x_{123} \\
R_{91} &= -x_{18}\,x_{125} + x_{128} \\
R_{92} &= x_{124} - x_{126}\,x_{128} - x_{127} \\
R_{93} &= -x_{128}\,x_{130} + x_{129} - x_{131} \\
R_{94} &= -x_{128}\,x_{133} + x_{132}\,x_{224} - x_{134} \\
R_{95} &= -x_{128}\,x_{136} + x_{135}\,x_{225} - x_{137} \\
R_{96} &= x_{138} - 2\,x_{138}\,x_{140} + x_{141} \\
R_{97} &= -x_{138}\,x_{141} - x_{141} + x_{142} \\
R_{98} &= x_{141}^2 - x_{142}^2 + x_{162} \\
R_{99} &= -x_{139}\,x_{144} - x_{144} + x_{145} \\
R_{100} &= 2\,x_{143} + x_{143}^2 + x_{144}^2 - x_{145}^2 \\
R_{101} &= 1 - 2\,x_{0} + x_{0}^2 - 2\,x_{0}\,x_{139} + x_{146} \\
R_{102} &= -1 - x_{0}\,x_{144} + x_{3} + x_{139}\,x_{144} - x_{143} + x_{144} + x_{146}\,x_{147} \\
R_{103} &= 1 - 2\,x_{5} + x_{5}^2 - 2\,x_{5}\,x_{139} + x_{148} \\
R_{104} &= -x_{5}\,x_{193} + x_{18} + x_{148}\,x_{149} - x_{192} + x_{194} \\
R_{105} &= -1 + 2\,x_{139}\,x_{208} - x_{150} + 2\,x_{208} - x_{208}^2 \\
R_{106} &= -1 + x_{98} + x_{150}\,x_{151} - x_{192} - x_{193}\,x_{208} + x_{194} \\
R_{107} &= -x_{139}\,x_{153} - x_{153} + x_{154} \\
R_{108} &= 2\,x_{152} + x_{152}^2 + x_{153}^2 - x_{154}^2 \\
R_{109} &= 1 - 2\,x_{98} + x_{98}^2 - 2\,x_{98}\,x_{139} + x_{155} \\
R_{110} &= -1 - x_{98}\,x_{153} + x_{99} - x_{152} + x_{154} + x_{155}\,x_{156} \\
R_{111} &= -1 - x_{98}\,x_{144} + 2\,x_{98}\,x_{157} + x_{100} - 2\,x_{139}\,x_{157} - x_{143} + x_{144} + x_{145} + x_{155}\,x_{157} - 3\,x_{157} \\
R_{112} &= 1 + x_{139} - x_{139}\,x_{159} - x_{159} + x_{160} \\
R_{113} &= 1 + 2\,x_{158} + x_{158}^2 - 2\,x_{159} + x_{159}^2 - x_{160}^2 \\
R_{114} &= -1 + x_{98} - x_{98}\,x_{159} + x_{104} + x_{155}\,x_{161} - x_{158} + x_{160} \\
R_{115} &= -2\,x_{139} - x_{139}^2 + x_{162} \\
R_{116} &= 1 - x_{98} - x_{98}\,x_{143} - x_{98}\,x_{145} + 2\,x_{98}\,x_{163} + x_{105} + x_{139}\,x_{145} - 2\,x_{139}\,x_{163} + x_{143} \\
&\quad - x_{144}\,x_{162} + 2\,x_{145} + x_{155}\,x_{163} - 3\,x_{163} \\
R_{117} &= -x_{139}\,x_{165} - x_{165} + x_{166} \\
R_{118} &= 2\,x_{164} + x_{164}^2 + x_{165}^2 - x_{166}^2 \\
R_{119} &= -1 - x_{98}\,x_{165} + x_{108} + x_{155}\,x_{167} - x_{164} + x_{166} \\
R_{120} &= -x_{139}\,x_{169} - x_{169} + x_{170}
\end{aligned}
$$

$$
\begin{aligned}
R_{121} &= 2\,x_{168} + x_{168}^2 + x_{169}^2 - x_{170}^2 \\
R_{122} &= -1 - x_{98}\,x_{169} + x_{111} + x_{155}\,x_{171} - x_{168} + x_{170} \\
R_{123} &= -x_{139}\,x_{173} - x_{173} + x_{174} \\
R_{124} &= 2\,x_{172} + x_{172}^2 + x_{173}^2 - x_{174}^2 \\
R_{125} &= -1 - x_{98}\,x_{173} + x_{114} + x_{155}\,x_{175} - x_{172} + x_{174} \\
R_{126} &= -x_{139}\,x_{177} - x_{177} + x_{178} \\
R_{127} &= 2\,x_{176} + x_{176}^2 + x_{177}^2 - x_{178}^2 \\
R_{128} &= -1 - x_{98}\,x_{177} + x_{117} + x_{155}\,x_{179} - x_{176} + x_{178} \\
R_{129} &= -x_{139}\,x_{181} - x_{181} + x_{182} \\
R_{130} &= 2\,x_{180} + x_{180}^2 + x_{181}^2 - x_{182}^2 \\
R_{131} &= -1 - x_{98}\,x_{181} + 2\,x_{98}\,x_{183} + x_{118} - 2\,x_{139}\,x_{183} + x_{155}\,x_{183} - x_{180} + x_{181} + x_{182} - 3\,x_{183} \\
R_{132} &= -x_{139}\,x_{185} - x_{185} + x_{186} \\
R_{133} &= 2\,x_{184} + x_{184}^2 + x_{185}^2 - x_{186}^2 \\
R_{134} &= -1 - x_{98}\,x_{185} + x_{121} + x_{155}\,x_{187} - x_{184} + x_{186} \\
R_{135} &= -x_{139}\,x_{189} - x_{189} + x_{190} \\
R_{136} &= 2\,x_{188} + x_{188}^2 + x_{189}^2 - x_{190}^2 \\
R_{137} &= -1 - x_{98}\,x_{189} + x_{124} + x_{155}\,x_{191} - x_{188} + x_{190} \\
R_{138} &= 1 - x_{139}\,x_{193} - x_{193} + x_{194} \\
R_{139} &= -1 + 2\,x_{192} + x_{192}^2 + x_{193}^2 - 2\,x_{194} - x_{194}^2 \\
R_{140} &= -x_{98}\,x_{193} + 2\,x_{98}\,x_{195} + x_{125} - 2\,x_{139}\,x_{195} + x_{155}\,x_{195} - x_{192} + x_{193} + x_{194} - 3\,x_{195} \\
R_{141} &= -x_{139}\,x_{197} - x_{197} + x_{198} \\
R_{142} &= 2\,x_{196} + x_{196}^2 + x_{197}^2 - x_{198}^2 \\
R_{143} &= -1 - x_{98}\,x_{197} + x_{129} + x_{155}\,x_{199} - x_{196} + x_{198} \\
R_{144} &= -x_{139}\,x_{201} - x_{201} + x_{202} \\
R_{145} &= 2\,x_{200} + x_{200}^2 + x_{201}^2 - x_{202}^2 \\
R_{146} &= -1 - x_{98}\,x_{201} + x_{132} + x_{155}\,x_{203} - x_{200} + x_{202} \\
R_{147} &= -x_{139}\,x_{205} - x_{205} + x_{206} \\
R_{148} &= 2\,x_{204} + x_{204}^2 + x_{205}^2 - x_{206}^2 \\
R_{149} &= -1 - x_{98}\,x_{205} + x_{135} + x_{155}\,x_{207} - x_{204} + x_{206} \\
R_{150} &= -4 - 2\,x_0 - 4\,x_0^2 - 2\,x_0\,x_4 - x_1^2 - x_2^2 - x_3^2 - x_4^2 - x_5^2 + 2\,x_9 + 2\,x_9\,x_{10} \\
&\quad - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 - x_{17}^2 - x_{18}^2 \\
&\quad - x_{20} - x_{96}^2 + 2\,x_{98} - 2\,x_{98}^2 - x_{99}^2 - x_{100}^2 - x_{104}^2 - x_{105}^2 \\
&\quad - x_{108}^2 - x_{111}^2 - x_{114}^2 - x_{117}^2 - x_{118}^2 - x_{121}^2 - x_{124}^2 \\
&\quad - x_{125}^2 - x_{129}^2 - x_{132}^2 - x_{135}^2 + x_{138} - x_{208}^2 \\
R_{151} &= -x_5 - x_5\,x_9 - x_5\,x_{10} - x_5\,x_{11} - x_5\,x_{12} - x_5\,x_{13} - x_5\,x_{14} - x_5\,x_{15} - x_5\,x_{16} \\
&\quad - x_5\,x_{17} + x_{208} \\
R_{152} &= -16 + x_0 - x_{209}^2 \\
R_{153} &= -x_{139}\,x_{211} - x_{211} + x_{220} \\
R_{154} &= -1 + x_{210}^2 + x_{211}^2 - x_{220}^2 \\
R_{155} &= -x_{213} - x_{213}\,x_{214} + x_{221} \\
R_{156} &= -1 + x_{212}^2 + x_{213}^2 - x_{221}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{157} &= -x_{205}^2 + x_{222} \\
R_{158} &= x_{211} - 2\,x_{216}\,x_{222} - x_{222} \\
R_{159} &= -4\,x_{211} - 8\,x_{211}\,x_{215} + x_{214} \\
R_{160} &= -x_{139} - x_{210}\,x_{217} + x_{214} \\
R_{161} &= -1 - x_{204} - x_{210}\,x_{218} + x_{212} \\
R_{162} &= -x_{96} - 4\,x_{205}\,x_{219} + x_{213} \\
R_{163} &= -2 + 2\,x_0^2 + 2\,x_0\,x_4 - 2\,x_0\,x_{144} - 2\,x_0\,x_{153} + x_1^2 - 2\,x_1\,x_{193} + x_2^2 - 2\,x_2\,x_{181} \\
&\quad + x_4^2 - 2\,x_4\,x_{153} + x_9^2 - 2\,x_9\,x_{159} + x_{10}^2 - 2\,x_{10}\,x_{165} + x_{11}^2 - 2\,x_{11}\,x_{169} \\
&\quad + x_{12}^2 - 2\,x_{12}\,x_{173} + x_{13}^2 - 2\,x_{13}\,x_{177} + x_{14}^2 - 2\,x_{14}\,x_{185} + x_{15}^2 \\
&\quad - 2\,x_{15}\,x_{189} + x_{16}^2 - 2\,x_{16}\,x_{197} + x_{17}^2 - 2\,x_{17}\,x_{201} - 2\,x_{96}\,x_{205} - x_{139}\,x_{223} \\
&\quad + x_{143}^2 + x_{152}^2 + x_{158}^2 + 2\,x_{159} + x_{164}^2 + x_{168}^2 + x_{172}^2 \\
&\quad + x_{176}^2 + x_{180}^2 + x_{184}^2 + x_{188}^2 + x_{192}^2 + x_{196}^2 + x_{200}^2 \\
&\quad + x_{204}^2 + 2\,x_{205}^2 - x_{227}^2 - x_{228}^2 - x_{229} - x_{229}^2 \\
R_{164} &= -x_{100}\,x_{224} + x_{125} \\
R_{165} &= -x_{105}\,x_{225} + x_{125} \\
R_{166} &= -x_{118}\,x_{226} + x_{125} \\
R_{167} &= -1 + x_2 - x_3\,x_{234} \\
R_{168} &= -2 - x_{96}^2 - x_{144}^2 - x_{153}^2 + 2\,x_{159} - x_{159}^2 - x_{165}^2 - x_{169}^2 \\
&\quad - x_{173}^2 - x_{177}^2 - x_{181}^2 - x_{185}^2 - x_{189}^2 - x_{193}^2 - x_{197}^2 \\
&\quad - x_{201}^2 + x_{205}^2 - x_{227}^2 - x_{228}^2 - x_{229} - x_{229}^2 \\
R_{169} &= -1 - 4\,x_{102}^2 - 4\,x_{107}^2 - 4\,x_{110}^2 - 4\,x_{113}^2 - 4\,x_{116}^2 - 4\,x_{120}^2 \\
&\quad - 4\,x_{123}^2 + x_{125}^2 - 4\,x_{127}^2 - 4\,x_{131}^2 - 4\,x_{134}^2 - 4\,x_{137}^2 - x_{230}^2 \\
&\quad - x_{231}^2 - x_{232} - x_{232}^2 \\
R_{170} &= -2 + x_{98} - 4\,x_{233} - 4\,x_{233}^2 \\
R_{171} &= -1 + x_{234} - 4\,x_{235} - 4\,x_{235}^2
\end{aligned}
$$

B The compact fixed coefficient specification

The compact equation retains its native/K215 ancestry. The direct quartics in Appendix A use the separate proof-calculus route. The four coefficient payloads in `compact/coefficients/` have bit lengths 590,069, 592,884, 590,069, and 590,070 for β, K_0, M_C, M_F respectively. The following complete finite recipe is retained unchanged and its reconstruction is checked exactly. In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$
\begin{aligned}
\mathcal{E} = \{0, \ldots, 942\} &\cup \{983, \ldots, 992\} \cup \{1003, \ldots, 1011\} \\
&\cup \{1013, \ldots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.
\end{aligned}
$$

Then

$$\begin{aligned}
M_C &= B - 1 - \sum_{e\in\mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063},\\
M_F &= B - 1 + 4 + (2^{51}-2)(\rho^{4393}+\rho^{5495})\\
&\quad + \rho^{1013}G_{10}(\rho) + \rho^{1065}G_9(\rho) + \rho^{2195} + \rho^{3296}\\
&\quad + (2^{65}-1)\rho^{8792}G_{13}(\rho) + (2^{19}-1)\rho^{8805},\\
K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849}\\
&\quad + (\rho^{4393}+\rho^{5495})\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) &= (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i\\
&\quad + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x),\\
\mathcal{C} &= \{(10,1023,8),(18,1013,9),(27,1003,8),\\
&\quad (35,983,9),(44,1031,1),(45,1022,1),(46,1011,1),\\
&\quad (47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.
\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}
h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0\le j<430\},\\
\mathcal{L} = \{u_j : d_j = 0,\ 0\le j<430\},\\
r_q = 6 + |\mathcal{R}\cap\{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L}\cap\{0,\dots,q-1\}|.
\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0\le i\le 942$; all index ranges are fixed:

index i	tile $t_i=(w,n,e,s)$
$0,1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j=2q+b<430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0,h(215,b),0,h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R},b=0,1)$	$(r_q,b,0,h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L},b=0,1)$	$(0,b,l_q,h(q,b))$
$924+j\ (0\le j<4)$	$(0,j+430\lfloor j/2\rfloor,0,438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t=434,435,436)$	$(1,438,1,t)$
$934+3a+b-6\lfloor a/4\rfloor$ $a\in\{0,1,4\},\ b=0,1,2$	$(a,437,b,437)$

For $t=(w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$.

For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \dots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \dots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
OOe35uwDf15DpT6VJWZBjSbHPpPAINYTteOeUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

C Exact compact degree certificates

Set $q_0 = \beta J$ and $k_0 = \eta + \zeta$. The complete highest homogeneous part of the principal equation is

$$[P_{\mathrm{tr}}]_{232} = -9\,2^{10} q_0^{100} w^{26} s^{58} i^{10} k_0^{32} (k_0 s T - q_0^2 (q_0 - F))^2.$$

It is nonzero because $\beta > 0$. The new degrees of f, V, Q, G, M, Z are respectively $36, 40, 80, 50, 11, 57$. Thus $-QM^2GV^2$ has degree $80 + 22 + 50 + 80 = 232$, while QZ^2 has degree 194 and the remaining terms are lower. The verifier computes the full highest forms, including every possible cancellation in the arithmetic circuit.

For each circuit, the following is one coefficient-certified term of its highest homogeneous part. Here β is a fixed positive integer and is assigned degree zero.

$$\begin{aligned}
d = 402: &\quad 1024\, J^{244} T^4 \beta^{244} \eta^{40} i^{12} s^{70} w^{32} \\
d = 352: &\quad -256\, J^{216} T^2 \beta^{216} \eta^{34} i^{10} s^{62} w^{28} \\
d = 290: &\quad -256\, J^{158} T^2 \beta^{158} \eta^{34} i^{10} s^{60} w^{26} \\
d = 258: &\quad -256\, J^{132} \beta^{132} \eta^{32} i^{10} s^{58} w^{26} \\
d = 232: &\quad -9216\, J^{106} \beta^{106} \eta^{32} i^{10} s^{58} w^{26}
\end{aligned}$$

References

[1] ProveIt contributors, *DiophantineConsistency research collection*, commit `b7e35224c04fe1031c0a53f80a395e9b32205022`, 25 October 2026. Pinned research collection on GitHub.

[2] AI-assisted ProveIt research, *Lucas corrections and witness elimination*, incoming archive dated 25 October 2026; exact bytes, proof, and coefficient closure retained under `compact/`.

[3] AI-assisted ProveIt research, *Removing factorial authenticity: binomial prime-power certificates for Diophantine consistency*, incoming archive dated 25 October 2026; source and proof under `direct/`.

[4] AI-assisted ProveIt research, *Shared factorial error*, incoming archive dated 25 October 2026; source and proof under `direct/`.

[5] B. Farhi, *An identity involving the least common multiple of binomial coefficients and its application*, American Mathematical Monthly **116** (2009), 836–839. https://arxiv.org/abs/0906.2295.